

AP CSP | Lesson 10: Boolean Logic

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: division gives 2.8 and $14 = 2 \times 5 + 4$. A truncates division. B uses the whole-group count for MOD. D reverses or confuses the two results.

Q2. Answer: D

D includes equality. A and C are strict comparisons and fail when the values are equal. B is false because the values are equal.

Q3. Answer: A

A is correct: passed stores true when assigned and is never reassigned. B wrongly recalculates after score changes. C confuses two variables. D treats assignment as a live formula.

Q4. Answer: B

B matches inclusive OR. A describes an exclusive choice, not AP OR. C invents numeric counting. D confuses a Boolean result with a string.

Q5. Answer: B and D

B reverses false. D is the correct De Morgan equivalence. A is false because AND needs both true. C needs AND between the negated operands; $A=\text{true}, B=\text{false}$ is a counterexample.

Q6. Answer: C

C enforces both inclusive bounds. A excludes the endpoints. B accepts out-of-range ages, such as 5. D accepts only the two endpoints, excluding valid interior ages.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) false. (b) true. (c) false. (d) true.

1 mark: (a) false.

1 mark: (b) true.

1 mark: (c) false.

1 mark: (d) true.

Q8. 4 marks

Answer: (a) true, true. (b) false, true. (c) false, true. (d) false, false.

1 mark: Both outputs correct for row (a): true, true.

1 mark: Both outputs correct for row (b): false, true.

1 mark: Both outputs correct for row (c): false, true.

1 mark: Both outputs correct for row (d): false, false.

Q9. 4 marks

Answer: (a) $(x \geq 4) \text{ AND } (x \leq 9)$. (b) $(x < 4) \text{ OR } (x > 9)$. (c) true. (d) false.

1 mark: Correct inclusive-range expression, or an equivalent expression.

1 mark: Correct outside-range expression with strict bounds and no NOT.

1 mark: At $x=4$ the inclusive-range result is true.

1 mark: At $x=10$ the inclusive-range result is false.

Q10. 4 marks

Answer: (a) $(\text{NOT } A) \text{ OR } (\text{NOT } B)$. (b) $(\text{NOT } A) \text{ AND } (\text{NOT } B)$. (c) true. (d) false; these two expressions are not equivalent.

1 mark: (a) Both operands negated and joined with OR.

1 mark: (b) Both operands negated and joined with AND.

1 mark: (c) true.

1 mark: (d) false.

Q11. 4 marks

Answer: (a) true. (b) true. (c) true. (d) false. The new run recomputes the conditions.

1 mark: (a) inRange is true.

1 mark: (b) hasEntry is true, including the both-true OR case.

1 mark: (c) eligible is true.

1 mark: (d) eligible is false after all conditions are recomputed for age=18.

Q12. 4 marks

Answer: (a) true. (b) $\text{inRange} \leftarrow (\text{age} \geq 12) \text{ AND } (\text{age} \leq 17)$. (c) true. (d) false.

1 mark: Buggy age=5 result is true.

1 mark: Correct assignment enforces both inclusive bounds with AND, or equivalent logic.

1 mark: Corrected result at age=12 is true.

1 mark: Corrected result at age=11 is false.